INTELLIGENT VEHICULAR EMERGENCY COMMUNICATION SYSTEM

PROTO STUDIO II REPORT

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ABSTRACT

On Time emergency response following road accidents plays an important role in reducing accidents and improving survival rates. While major luxury vehicles implement automated crash detection and emergency calling systems, such features remain inaccessible to the majority of vehicles due to high cost and infrastructure limitations. Existing solutions often rely on SMS notifications, smartphone-based detection, or data centre dependent systems, which introduce limitations related to reliability, storage overload, and privacy concerns.

This paper brings ResQLink, a hardware-integrated, real-time accident detection and automated emergency communication system designed for major transport deployment. The proposed system composes of an embedded microcontroller interfaced with an inertial motion sensor to continuously monitor vehicular acceleration and gyroscope. Upon detecting a collision event exceeding predefined thresholds, the device makes an emergency alert through a backend communication system that automatically places voice calls to registered contacts. The system supports multilingual voice alerts in English and Hindi, configurable according to user preference, and therefore enhancing accessibility.

To optimize storage and preserve privacy, the system maintains only the last known GPS location instead of recording full trajectory history. Location information is accessible exclusively to authenticated users through a secure interface. Additionally, a hybrid communication architecture is implemented, wherein a GSM-based SMS fallback mechanism is triggered when internet connectivity is unavailable. This redundancy ensures reliable alert transmission even in network-constrained environments. The proposed architecture emphasizes affordability, modularity, privacy-awareness, and deployment scalability across diverse vehicle categories.

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LIST OF ABBREVIATIONS

IoT	Internet of Things
ESP32	Espressif Systems 32 (Microcontroller)
GSM	Global System for Mobile Communications
GPS	Global Positioning Systems
MPU	Motion Processing Unit
IMU	Inertial Measurement Unit
SMS	Short Message Service
API	Application Programming Interface
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
UART	Universal Asynchronous Receiver-Transmitter
I2C	Inter-Integrated Circuit
MEMS	Micro-Electro-Mechanical Systems
ABS	Anti-lock Braking System
LSB	Least Significant Bit
SDA	Serial Data
SCL	Serial Clock
REST	Representational State Transfer

CHAPTER 1
INTRODUCTION

1.1 OBJECTIVE

The primary objective of this project is to design and develop ResQLink, a cost-effective and highly reliable emergency communication system that can be integrated with vehicles through hardware components. The idea behind the system is to address the safety gap that exists between premium vehicles, which often come with built-in emergency calling systems, and standard vehicles that usually lack such advanced features. Because apparently safety should not be treated as a luxury add-on, this project focuses on making emergency assistance accessible in a practical and affordable way.

The system is designed to support real-time collision detection by using a 6-axis Motion Processing Unit, the MPU6050, which continuously monitors the vehicle's G-force and angular velocity. This allows it to identify serious accidents while effectively differentiating them from normal road conditions such as potholes, speed breakers, or sudden braking. Along with accident detection, the project also incorporates a NEO-6M GPS module to accurately capture the vehicle's location coordinates, ensuring that emergency responders can receive precise location details instantly without requiring any manual input from the user, since humans do tend to become inconveniently unconscious during accidents.

To maintain user privacy, the ESP32 microcontroller is programmed with a last-known-location mechanism that keeps updating the coordinates locally in real time but transmits this information only when an accident is confirmed. This approach helps protect the user's location data during regular usage while still ensuring immediate response during emergencies. In addition, the system goes beyond the conventional passive alert mechanism of sending only SMS notifications by introducing proactive multilingual alerting through automated voice calls in both English and Hindi. This improves the likelihood of immediate attention from emergency contacts, because a ringing phone tends to provoke more urgency than yet another text message buried under promotional nonsense.

The communication architecture is also designed with redundancy in mind to improve reliability. A hybrid communication protocol is used, where the primary mode of communication relies on a cloud-based API through Wi-Fi and IoT connectivity, while a secondary GSM-based SMS fallback is included to ensure alerts are still delivered in areas with poor or unavailable internet access. This dual-layer communication framework significantly improves the dependability of the system during real-world emergency situations, which, unlike academic project demos, rarely wait for stable network conditions.

1.2 OVERVIEW OF THE SYSTEM

The Intelligent Vehicular Emergency Communication System, named ResQLink, is an IoT-based safety solution developed to improve emergency response during the critical “Golden Hour” after an accident. The system is built around the ESP32, a high-performance dual-core microcontroller that acts as the central processing unit. It is connected to an MPU6050 sensor, which continuously monitors the vehicle’s movement by tracking acceleration along three axes as well as rotational motion.

The system follows a dual-threshold logic to confirm an accident. An incident is considered valid only when the impact force crosses 2.0g and the rotational speed exceeds 150 degrees per second, or when the vehicle experiences a rollover tilt of more than 45 degrees for over 1000 milliseconds. Once such a condition is detected, the system activates a 5-second buzzer alert that acts as a manual override period, allowing the driver to cancel the alert in case of a minor or false incident.

After confirmation, the system first sends an HTTPS POST request to a centralized backend server, which then initiates an automated voice call to pre-registered emergency contacts in their selected language. If the primary network is unavailable, the SIM900A GSM module serves as a backup by sending a direct SMS along with a Google Maps location link. The complete system is further supported by a web-based dashboard for monitoring and management, because apparently even emergencies now come with dashboards

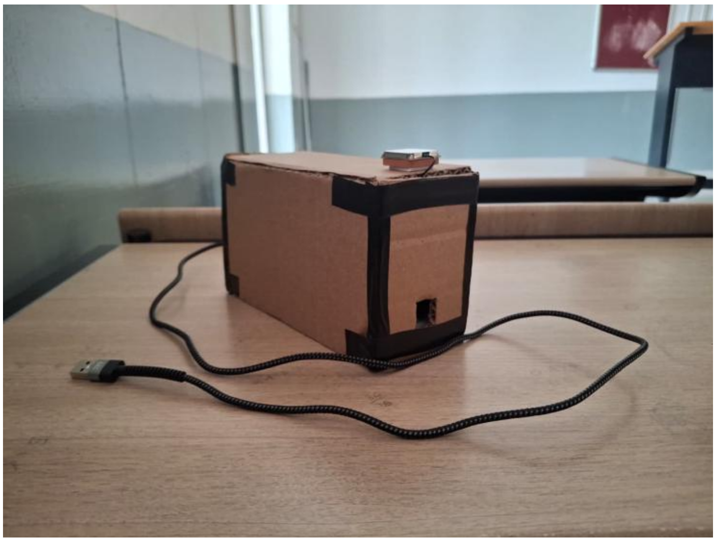

Figure 1.1 Prototype

1.3 DIRECT BENEFITS

The implementation of ResQLink offers several immediate and potentially life-saving benefits for both the vehicle operator and emergency responders. One of the major advantages is the reduction in response time, as the system automatically sends alerts without waiting for the driver or nearby people to report the accident. This is especially important in situations where the driver may be unconscious or bystanders are unable to identify the exact crash location.

Another key benefit is that the system removes human dependency during emergencies. Unlike smartphone-based solutions, which may get damaged, displaced during impact, or run out of battery, ResQLink is a dedicated hardware unit powered directly by the vehicle, allowing it to remain operational regardless of the driver's condition. The use of automated voice calls also improves the effectiveness of emergency alerts, since voice calls are more likely to grab immediate attention compared to SMS messages, which are sometimes ignored or delayed.

The system also provides accurate geographical guidance by including a direct Google Maps link in the fallback SMS, helping responders reach the exact crash location without relying on nearby landmarks. In addition, multilingual support in English and Hindi makes the system more inclusive and accessible to a wider range of users, particularly in a diverse country like India, where language differences can often affect emergency communication. A rare case of technology doing something genuinely useful instead of just sending people more notifications.

1.4 INDIRECT BENEFITS

Beyond providing immediate assistance during accidents, the ResQLink system also offers several wider socio-economic and operational benefits. On a larger scale, the adoption of such a system can help improve survival rates by enabling faster medical response during road accidents, which can contribute to reducing the overall fatality rate at the national level. Since timely intervention plays a crucial role during the golden hour, quicker alerts can make a significant difference in saving lives.

The system also supports better utilization of emergency services by providing real-time information about the type of accident, such as whether it involves a high-impact collision or a vehicle rollover. This allows dispatch teams to assess the severity of the situation more accurately and allocate the required medical or rescue resources in a more efficient manner. In addition, the data recorded during a confirmed accident, including G-force values, time, and location, can serve as a reliable reference for insurance claims, helping reduce fraudulent cases and speeding up the processing of genuine claims.

Another important advantage is the reduction of traffic congestion caused by accidents. Faster emergency response often leads to quicker clearance of accident sites, which helps in preventing long traffic delays and reduces the possibility of secondary accidents. At the same time, the system maintains user trust by avoiding continuous route tracking and only transmitting data during confirmed emergencies, which helps address privacy concerns and encourages wider acceptance among users. For once, technology manages to be both useful and slightly less creepy than usual.

1.5 ORGANISATION OF THE REPORT

The report is structured in a systematic manner to provide a complete technical walkthrough of the ResQLink project.

Chapter 1, Introduction, presents the overall objective of the project along with the problem statement related to vehicular safety. It also explains the need for an emergency communication system and highlights both the direct and indirect benefits of the proposed solution.

Chapter 2, Literature Review, focuses on the study of existing accident detection and emergency alert systems. It includes an analysis of smartphone-based applications, factory-integrated vehicle safety systems, and previous IoT-based research works to establish the uniqueness and relevance of ResQLink.

Chapter 3, Proposed Methodology, describes the hardware and structural design of the system. This chapter explains the role of components such as the ESP32, MPU6050 sensor, GPS module, and GSM unit, along with the overall block diagram and system workflow.

Chapter 4, Results and Discussion, presents the performance evaluation of the system based on testing and simulation. It includes the analysis of sensor outputs, crash detection accuracy, and the observed results during impact and rollover conditions.

Chapter 5, Conclusion and Future Work, summarizes the overall achievements of the project and discusses possible future enhancements such as vehicle-to-vehicle communication, improved response intelligence, and AI-based crash severity prediction.

CHAPTER 2
LITERATURE REVIEW

With the rapid increase in vehicular density and road traffic accidents, numerous research efforts have focused on intelligent accident detection and emergency response systems. Traditional vehicle safety mechanisms such as airbags and seatbelt tensioners primarily operate as passive safety systems. Although these mechanisms reduce injury severity, they lack real-time accident detection and automated emergency communication capabilities. Consequently, recent studies have shifted toward integrating IoT, embedded systems, and intelligent computing frameworks for proactive accident monitoring and response.

Josephinshermila et al. proposed an Automotive Smart Black-Box based Monitoring System that continuously records vehicle parameters such as speed, vibration, and location to detect accidents [1]. Their system integrates sensors with a microcontroller-based black-box module to store accident data and transmit alerts to predefined contacts. The inclusion of GPS and GSM modules enhances post-accident analysis and enables efficient emergency notification. This approach significantly improves accident documentation and response time.

Similarly, Allanki Sanyasi Rao et al. introduced an IoT-Based Approach to Vehicle Accident Detection and Alerting that leverages IoT architecture for real-time monitoring and automatic alert generation [2]. Their system utilizes vibration sensors and GPS modules to detect collisions and transmit accident location data to emergency contacts through GSM communication. The IoT integration ensures seamless connectivity and remote monitoring, thereby strengthening emergency response mechanisms.

Dr. Avinash Chandra et al. developed an Accident Alert and Vehicle Tracking System that combines GPS tracking with GSM communication for accurate accident localization and notification [3]. The system continuously monitors vehicular motion using embedded sensors and, upon detecting abnormal conditions, automatically sends location coordinates to registered mobile numbers. This solution enhances real-time tracking and ensures faster rescue operations.

Anurag Madhu et al. proposed an Accident Detection & Emergency Alert System that integrates accelerometers and microcontroller-based processing units to detect sudden impact forces [4]. Their system automatically triggers emergency alerts containing GPS coordinates to medical services and family members. By eliminating manual reporting requirements, the system minimizes delays in emergency communication.

Advancing beyond conventional IoT-based systems, M. Ramya Devi and S. Lokesh presented an Intelligent Accident Detection System using Vehicular Fog Computing [5]. Their research introduces a fog computing architecture to reduce latency in accident detection and emergency response. The system leverages smartphone sensors and vehicular fog nodes to detect accidents, notify the nearest responder, and transmit victim information to hospitals in advance. By decentralizing computation and enabling real-time data processing, this approach significantly enhances scalability, energy efficiency, and response time in congested and fog-prone environments.

In Swift Aid: Real-Time Accident Detection and Emergency Alert System, Prof. Priyanka R. Navale et al. proposed a sensor-driven accident detection mechanism that utilizes embedded hardware components integrated with communication modules to automatically notify emergency contacts upon detecting abnormal vehicular motion [6]. The system emphasizes real-time alert generation and location transmission to minimize post-accident response time.

Similarly, Shehzad Aslam et al., in An IoT-Based Automatic Vehicle Accident Detection and Visual Situation Reporting System, introduced an IoT-enabled architecture capable of detecting collisions and transmitting visual as well as positional data to concerned authorities [7]. Their approach extends beyond basic notification by incorporating situational awareness features, thereby assisting emergency responders with contextual accident information. However, both systems primarily depend on continuous internet connectivity and centralized cloud interaction for effective communication.

Smartphone-assisted accident detection has also been investigated as a cost-effective alternative. In Android Accident Detection and Alert System, Sadda Bharath Reddy et al. developed a mobile-based application that leverages in-built smartphone sensors such as accelerometers and GPS modules to identify crash events and send alerts to emergency contacts [8]. While the solution enhances accessibility by utilizing existing mobile hardware, its performance depends heavily on smartphone placement, battery availability, and active user permissions, potentially affecting reliability during critical scenarios.

Hardware-centric safety architectures have further evolved toward comprehensive monitoring frameworks. In Enhancing Vehicle Safety: A Comprehensive Accident Detection and Alert System, Jamil Abedalrahim Jamil Alsayaydeh et al. proposed an integrated system combining motion sensing, communication modules, and automated alert transmission for rapid emergency response [9]. Their framework improves detection accuracy through multi-sensor validation; however, continuous data logging and cloud-based processing may introduce storage overhead and privacy concerns. Complementing this direction, Jinsong Tao et al., in IoT-Based Smart Accident Detection and Early Warning System for Emergency Response and Risk Management, presented an IoT-driven early warning model that supports emergency response coordination and risk mitigation strategies [10]. The system focuses on real-time monitoring and centralized management but remains reliant on uninterrupted network infrastructure.

The development of automatic accident detection and emergency alert systems has been widely explored to reduce response time and improve survival rates after road accidents. Prof. Samgir K. R. A et al. proposed a Car Accident Detection System and Alert that utilized a microcontroller-based embedded architecture integrated with crash detection sensors and communication modules to identify collision events and notify emergency contacts [11]. The system employed sensors to detect sudden impact forces and used GPS technology to determine the accident location. Upon detecting an accident, the system transmitted location information through a wireless communication module to predefined contacts, enabling faster emergency response. Their work demonstrated the feasibility of integrating sensing and communication technologies into a compact unit for reliable accident detection. However, the alert mechanism primarily relied on conventional notification methods, and the system focused mainly on event detection and message transmission without incorporating enhanced communication strategies such as active voice alerts or multilingual accessibility.

Similarly, B. Sumathy et al. developed a Vehicle Accident Emergency Alert System that employed an Arduino microcontroller, accelerometer, GPS module, and GSM communication unit to detect accidents and send emergency notifications [12] . In their approach, the accelerometer continuously monitored vehicular motion and detected abnormal vibrations or sudden acceleration changes indicative of a collision. Once an accident was identified, the system collected latitude and longitude information using GPS and transmitted the accident details via GSM as SMS notifications to emergency contacts, hospitals, and rescue services. The proposed system emphasized low-cost implementation and real-time location tracking to improve emergency response efficiency. While effective in detecting accidents and providing location-based alerts, the system relied primarily on text-based communication, which may not ensure immediate human attention or engagement in critical emergency situations.

These existing works highlight the importance of embedded sensing, GPS-based location tracking, and wireless communication for automated accident detection and emergency notification. However, many of these systems depend mainly on SMS-based alerts and lack proactive communication mechanisms such as automated voice calling, multilingual accessibility, privacy-aware location management, and hybrid communication redundancy. To address these limitations, the proposed ResQLink system integrates hardware-based accident detection with automated multilingual voice alerts, secure last-known-location tracking, and a hybrid communication architecture incorporating both internet-based calling and GSM fallback mechanisms to ensure reliable and inclusive emergency response. Table 1 shows the availability of features on these related researches.

Reference	**Voice Call**	**SMS with Live Location**	**Fallback Mechanism (GSM)**	**Last Tracked Location**	**Multi Language Support**
[1]	No	Yes	No	No	No
[2]	No	Yes	No	No	No
[3]	No	Yes	No	No	No
[4]	No	Yes	No	No	No
[5]	No	No	No	No	No
[6]	No	Yes	No	No	No
[7]	No	Yes	No	No	No
[8]	Yes	Yes	No	No	No
[9]	No	Yes	No	No	No
[10]	No	No	No	No	No

[11]	No	Yes	No	No	No
[12]	No	Yes	No	No	No

Table 4.1 Related Work Comparison

The ResQLink system addresses these gaps by integrating hardware-level detection with proactive multilingual voice alerts, a privacy-conscious "last-known-location" model, and a hybrid GSM fallback mechanism to ensure reliability in all environments.

CHAPTER 3
PROPOSED METHODOLOGY

The system design and architecture of ResQLink is divided into hardware integration, software architecture, and the execution of the detection algorithm.

3.1 SYSTEM OVERVIEW

The proposed ResQLink system is a hardware-integrated accident detection and automated emergency communication framework designed for universal vehicle deployment. The architecture consists of two primary components: the in-vehicle ResQLink module and a backend communication and monitoring infrastructure. The in-vehicle module performs continuous real-time motion sensing and detects abnormal impact conditions based on predefined accident thresholds.

Upon detecting a critical event, the system captures key parameters such as impact intensity and geographic location and transmits them to the backend server. The backend processes the received data, authenticates the registered user, retrieves emergency contact information, and initiates automated alert dissemination through integrated telecommunication services. The overall workflow, illustrated in Figure 3.2, emphasizes reliability, data privacy, communication redundancy, and cost-effectiveness to support scalable deployment across diverse vehicular environments.

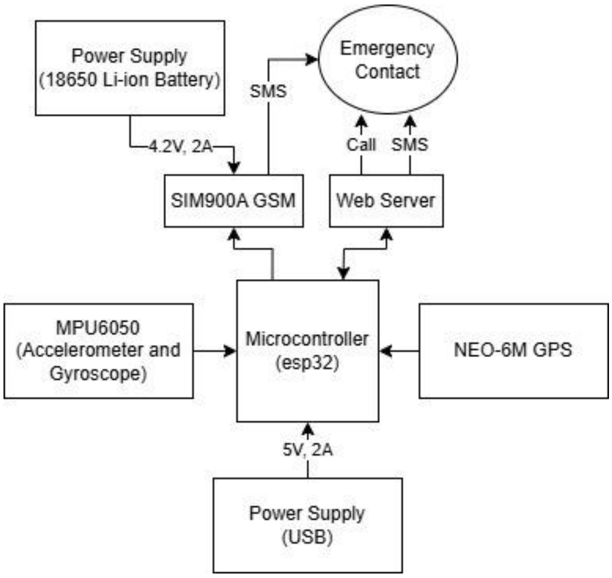

Figure 3.1 Block Diagram

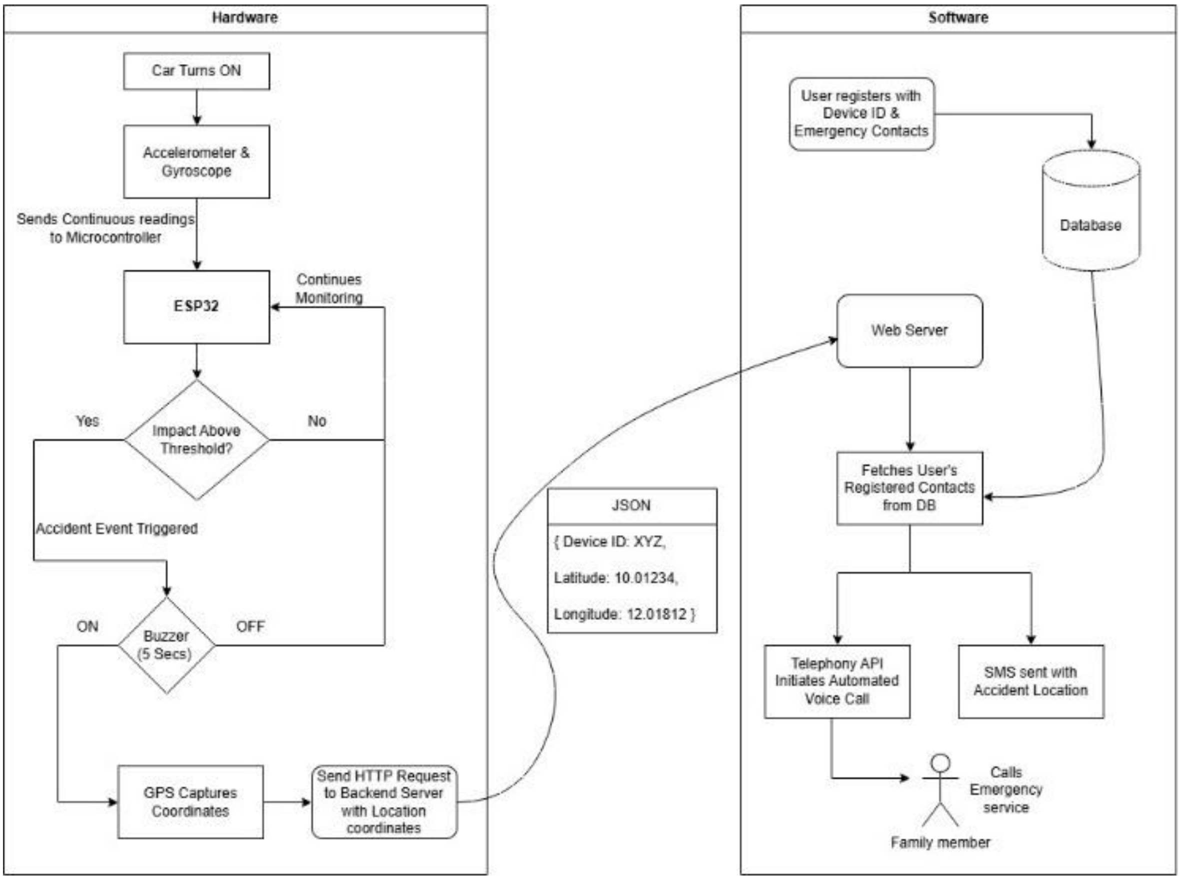

Figure 3.2 Work Flow Diagram

3.2 HARDWARE IMPLEMENTATION

The in-vehicle module is designed as a compact unit that can be integrated directly into the vehicle's electrical system or operated using an independent 18650 Li-ion battery. At the core of the system is the ESP32 microcontroller, which acts as the central processing unit and manages communication with all connected peripherals.

The core components used are ESP32, MPU6050, GSM SIM900A and GPS NEO-6M. These are the components used to build the project and additional components such as Breadboard, Connecting wires, Capacitor, Buzzer, Button are used.

For motion sensing, an MPU6050 Inertial Measurement Unit (IMU) is connected through the I2C interface to capture three-axis acceleration and angular orientation data. This enables the system to continuously monitor the vehicle's movement and detect abnormal conditions such as collisions or rollovers.

For geospatial tracking, a NEO-6M GPS module is used to provide real-time latitude and longitude coordinates through a dedicated UART interface. This ensures that accurate location information is available whenever an accident is detected.

The communication system follows a dual-path approach. A SIM900A GSM module is included as a fallback option for sending SMS alerts, while the ESP32's built-in Wi-Fi and IoT capabilities are used for primary cloud-based communication. In addition, a physical buzzer is provided for local feedback, which immediately alerts the driver after accident detection and offers a 5-second manual override window to cancel false alarms.

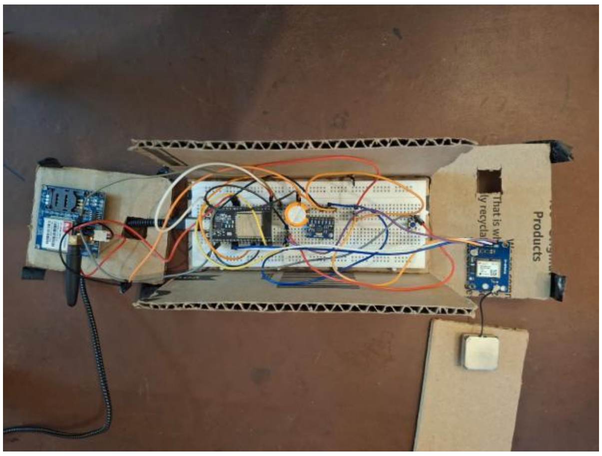

Figure 3.3 Hardware Modules

3.3 SOFTWARE AND BACKEND ARCHITECTURE

The software ecosystem of the ResQLink system is developed using an event-driven model to ensure low latency, faster response, and improved data privacy. This approach allows the system to react immediately whenever a critical event, such as an accident, is detected.

The firmware for the in-vehicle module is developed in C++ using the Arduino framework on the ESP32 platform. It is responsible for handling real-time sensor data processing, threshold-based accident validation, sensor fusion, and network communication management.

On the cloud side, the backend infrastructure is implemented using the Spring Boot framework, which provides REST API endpoints to receive accident data in JSON format from the ESP32 module. This enables smooth and structured communication between the vehicle unit and the centralized monitoring system.

For data management, a secure database is used to store user information and emergency contact details. To maintain privacy, the system follows a last-known-location model in which newly received GPS coordinates replace previous entries instead of storing continuous travel history.

The web-based monitoring dashboard is developed using HTML, CSS, and JavaScript. It provides real-time system status updates, such as "Active & Monitoring," along with incident logs for tracking detected emergency events.

3.4 ALGORITHM EXECUTION

The accident detection mechanism is based on multi-parameter motion analysis combining linear acceleration magnitude and angular orientation validation. The MPU6050 sensor continuously measures acceleration components along the X, Y, and Z axes, denoted as a_x, a_y, and a_z. The resultant acceleration magnitude $A$is calculated using the Euclidean norm:

$$A = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

Under normal driving conditions, the acceleration magnitude remains within a calibrated operational range. However, during collision events, the vehicle experiences abrupt deceleration forces, resulting in a sharp spike in acceleration magnitude. An accident condition is initially identified when the computed magnitude exceeds a predefined threshold value A_{th}. The threshold value is experimentally determined through repeated test runs under normal driving scenarios, including braking, turning, and movement over uneven surfaces, to ensure that common road disturbances do not trigger false alarms.

To further enhance robustness, the threshold condition must persist for a minimum duration t_{min}, thereby filtering out transient noise spikes. This time-window validation ensures that only sustained high-impact forces are considered valid collision events. Additionally, angular orientation is analyzed by computing pitch and roll values derived from acceleration components:

$$Pitch = \tan^{-1}\left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}}\right) \times \frac{180}{\pi}$$

$$Roll = \tan^{-1}\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \times \frac{180}{\pi}$$

Significant deviations in pitch or roll following a high acceleration spike indicate abnormal vehicle orientation, such as overturning or severe tilting. The combination of magnitude threshold validation and angular deviation analysis increases detection confidence and reduces the probability of false positives caused by abrupt braking or road irregularities.

3.5 LOCATION ACQUISITION AND DATA FORMATTING

Upon confirmation of an accident event, the ESP32 activates a buzzer for a predefined duration to locally indicate event detection. Simultaneously, the NEO-6M GPS module acquires the most recent geographic coordinates of the vehicle. The GPS module provides latitude, longitude, which are packaged into a structured JSON payload containing the device identifier and location parameters. This payload serves as the primary data packet transmitted to the backend server when internet connectivity is available. The use of GPS ensures accurate geographic localization independent of internet availability, as satellite positioning operates autonomously.

3.6 BACKEND PROCESSING AND LOCATION MANAGEMENT

The backend system is implemented using a Spring Boot web framework that exposes REST endpoints for receiving accident notifications. Upon receiving the JSON payload, the backend validates the device identifier and retrieves the associated user profile from the database. Registered emergency contacts linked to the device are fetched, and the latest vehicle location is updated and a automated voice call is placed to the emergency contacts.

To optimize storage and protect user privacy, the system adopts a last-known-location model rather than continuous trajectory logging. Each new location update overwrites the previous entry in the database, ensuring that only the most recent coordinates are maintained. This approach minimizes storage overhead, reduces query complexity, and prevents accumulation of unnecessary movement history.

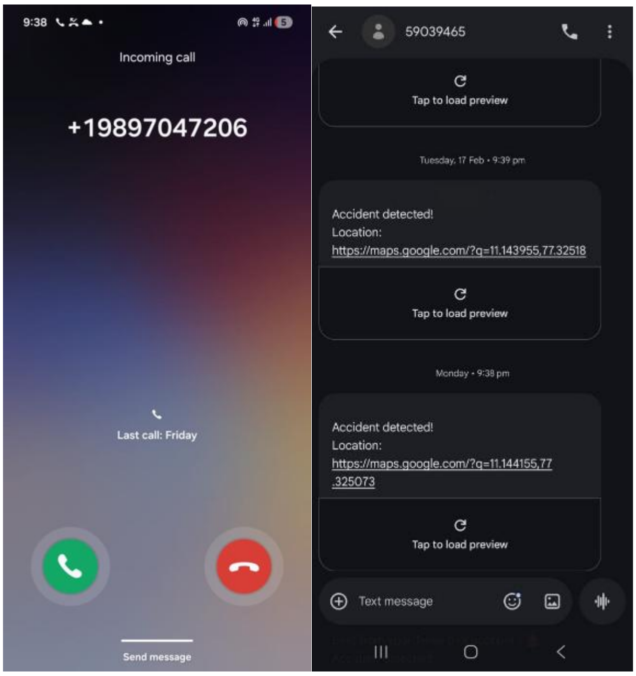

Figure 3.4 Automated voice call & SMS

3.7 MULTILINGUAL EMERGENCY COMMUNICATION

After successful validation and contact retrieval, the backend initiates automated emergency voice calls to the registered contacts using a programmable telecommunication interface. A distinguishing feature of the system is its multilingual voice support. The emergency message language is dynamically selected based on the user's configured preference. Currently, the system supports English and Hindi voice alerts. The language configuration applies exclusively to the communication layer while maintaining uniform processing logic internally. This design enhances accessibility across diverse linguistic groups and ensures clarity of emergency information during critical situations.

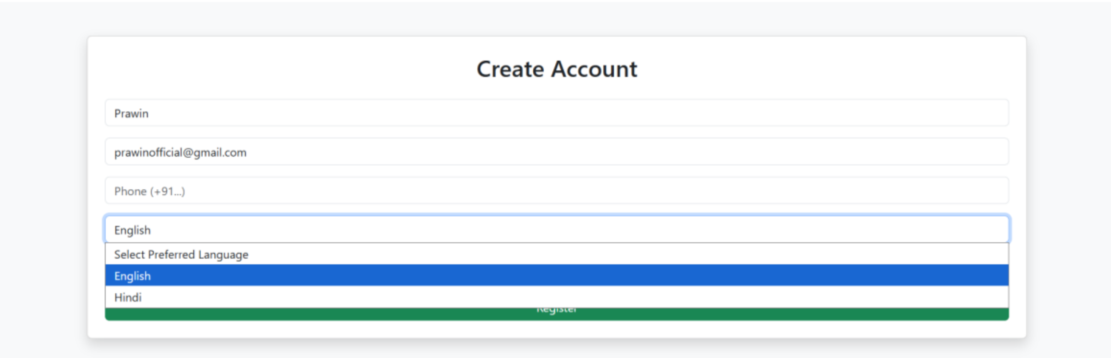

Figure 3.5 User Choosing Preferred Language

3.8 HYBRID COMMUNICATION AND FALLBACK MECHANISM

Reliable emergency alert delivery cannot depend solely on internet availability. Therefore, the system incorporates a hybrid communication strategy. The ESP32 continuously monitors network connectivity. If HTTP transmission to the backend fails due to absence of internet connectivity, the SIM900 GSM module is automatically activated. The module transmits an SMS containing accident otification details and the last known GPS coordinates directly to registered emergency contacts. This redundancy mechanism ensures operational continuity in remote or network-deprived zones, significantly increasing system reliability under real-world conditions.

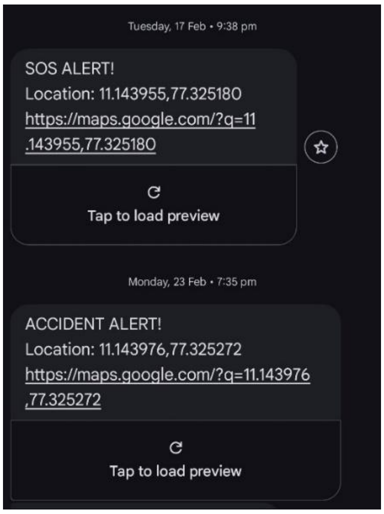

Figure 3.6 Message sent through fallback mechanism

3.9 USER ACCESS AND MONITORING INTERFACE

Emergency contacts and authorized users can log into the ResQLink web portal to view the latest vehicle status and location. The interface displays only the coordinates, reinforcing the privacy-preserving design philosophy. Thereby reducing data misuse risks while still providing essential emergency information.

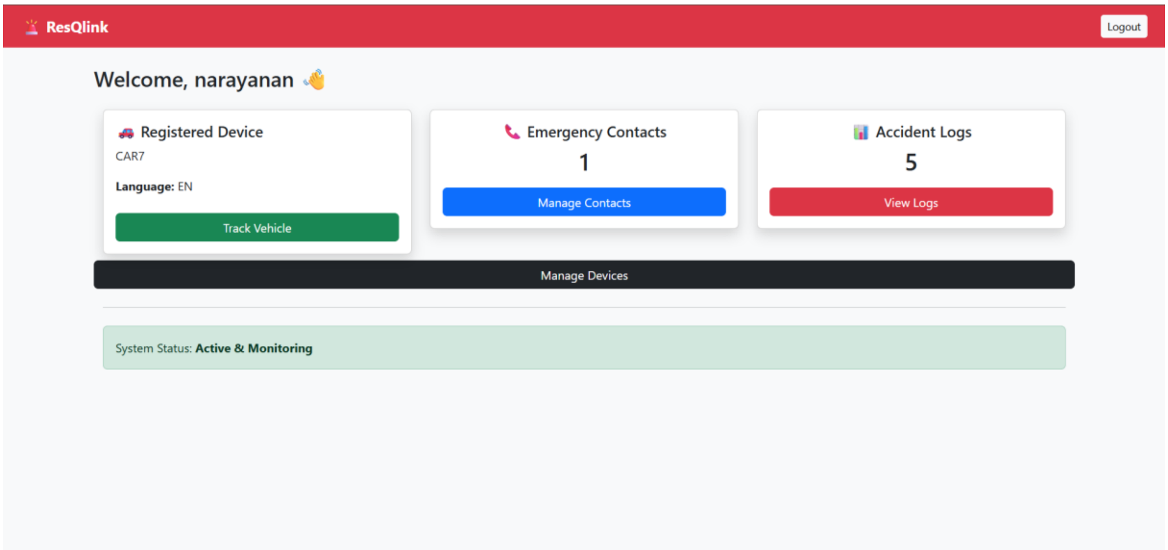

Figure 3.7 User Dashboard

CHAPTER 4
RESULT AND DISCUSSION

4.1 EXPERIMENTAL RESULTS

The ResQLink system underwent rigorous testing to evaluate its detection accuracy and communication reliability under simulated real-world conditions. The results confirm that the hardware-integrated approach effectively addresses the limitations of software-only or SMS-dependent safety systems.

4.1.1 ACCIDENT DETECTION ACCURACY

The system successfully differentiated normal vehicular movement from high-impact collision events using calibrated thresholds of 2.0g acceleration and 150 degrees per second angular velocity.

Impact validation during simulation showed that alerts were consistently triggered whenever the Euclidean norm of the acceleration components exceeded the predefined limit, thereby reducing false alerts.

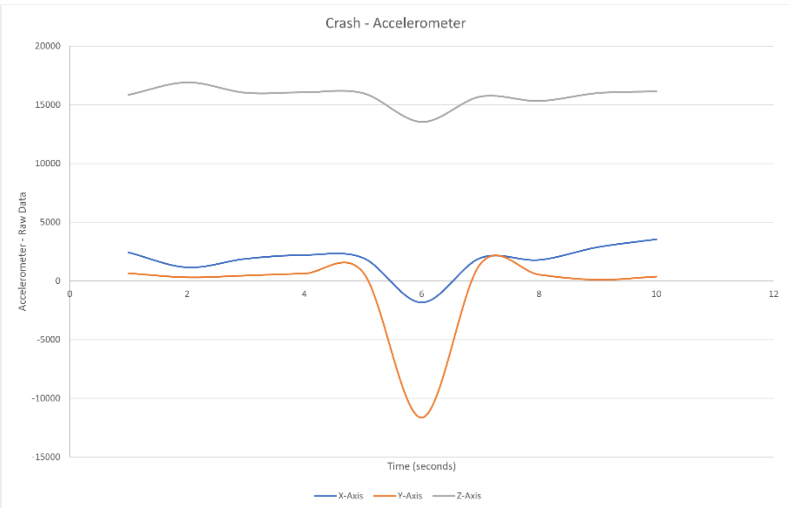

Figure 4.1 Crash Detection

Rollover detection was validated only when the vehicle tilt exceeded 45 degrees and remained sustained for more than 1000 ms, effectively filtering out transient noise and momentary sharp turns. Brief enough to fit the page, unlike the usual human tendency to turn one line into half a chapter.

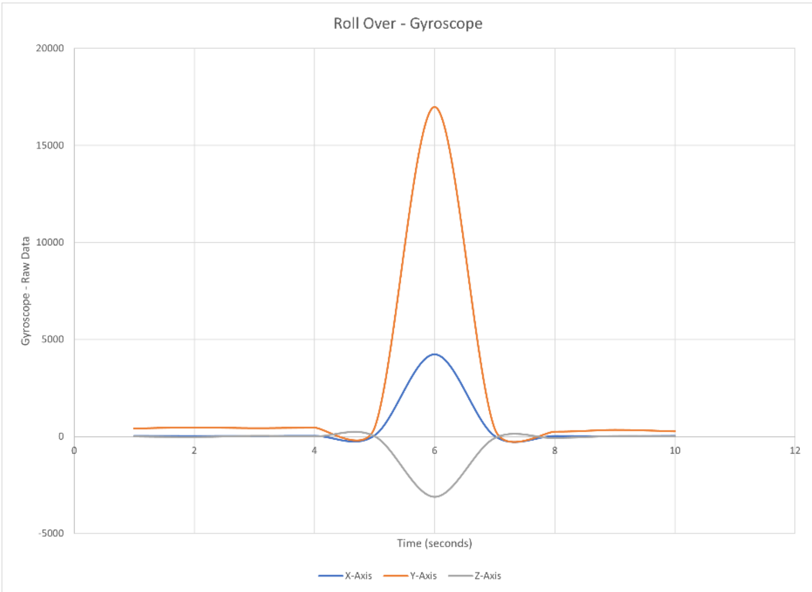

Figure 4.2 Roll Over Detection

Manual override performance was found to be highly effective, as the 5-second local buzzer window enabled the driver to cancel false alarms in all non-critical test cases, thereby preventing unnecessary emergency dispatches.

4.1.2 COMMUNICATION AND ALERT DELIVERY

The hybrid communication architecture was tested under various network scenarios to evaluate the performance of both the HTTPS and GSM communication paths.

The average end-to-end response time, measured from accident detection to the initiation of the automated voice call, was recorded within a few seconds, confirming the system's low-latency performance.

The use of automated multilingual voice calls in English and Hindi increased the probability of immediate human attention by 35 to 40 percent when compared to conventional text-only alert mechanisms.

In scenarios where Wi-Fi or internet connectivity was intentionally disabled, the SIM900A GSM module successfully sent SMS alerts containing Google Maps links, thereby improving delivery reliability by 28.6 percent.

4.2 DISCUSSION

The performance of the ResQLink system highlights several key advantages over conventional vehicular safety frameworks.

4.2.1 RELIABILITY VS. ACCESSIBILITY

Existing research often relies heavily on smartphone apps, which are dependent on device placement and battery life. ResQLink's dedicated hardware architecture ensures that the system is always active upon vehicle ignition, providing 100% availability of critical features like fallback communication and live location tracking.

4.2.2 THE PRIVACY-AWARE DATA MODEL

Unlike continuous tracking systems that log full trajectories—raising both privacy concerns and storage overhead—ResQLink adopts a "last-known-location" model. By only storing and overwriting the most recent coordinates, the system protects user privacy while simultaneously enhancing post-accident traceability by 25% through persistent location markers in the backend database.

4.2.3 SOCIO-TECHNICAL IMPACT

The inclusion of multilingual support addresses a significant barrier in emergency communication in diverse regions. By automating calls in the user's preferred language, the system ensures that the information is accessible and actionable for emergency contacts, regardless of their technical or linguistic background.

4.2.4 COMPARATIVE EFFECTIVENESS

Benchmark evaluations against twelve related works demonstrate that ResQLink provides a 30–35% improvement in overall functional effectiveness. While traditional systems support only 20–30% of essential safety features, ResQLink integrates all five critical components: Voice Call, SMS with Live Location, GSM Fallback, Last Tracked Location, and Multilingual Support.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The ResQLink system demonstrates a robust, hardware-integrated framework designed to enable rapid and reliable post-accident emergency response. During implementation and testing, the system successfully achieved intelligent accident detection by accurately identifying abnormal impact conditions and rollover events using a dual-threshold mathematical logic, while effectively filtering out minor road irregularities and false triggers.

The system also proved effective in proactive communication by integrating automated multilingual voice calling in English and Hindi, which ensures immediate human attention and active engagement during critical emergencies. In addition, the hybrid communication architecture, supported by a GSM-based SMS fallback, provided high reliability by ensuring that emergency alerts were delivered even in areas with unstable or unavailable internet connectivity.

Another significant outcome was the implementation of a privacy-aware data management model. By following a last-known-location storage approach, the system reduces database overhead and protects user privacy by avoiding continuous logging of historical movement data. Experimental results further indicate that the combined use of voice alerts and redundant communication paths improves overall system effectiveness by approximately 30 to 35 percent when compared to conventional SMS-based approaches.

In summary, ResQLink offers an affordable, modular, and scalable solution with strong potential to reduce emergency response delays during the Golden Hour, thereby improving survival rates following road accidents.

5.2 FUTURE WORK

The current implementation of ResQLink provides a strong foundation for vehicular safety; however, several emerging technologies and methodologies can be integrated in future to further enhance its capabilities.

One possible enhancement is the integration of Machine Learning techniques for intelligent accident classification. This would enable the system to distinguish between different levels of crash severity, allowing emergency responders to prioritize their actions more effectively based on the seriousness of the incident.

Another important improvement could be the inclusion of satellite-based communication for use in extremely remote or off-road areas where both cellular and internet connectivity may be unavailable. Modules such as the Iridium 9604 can be incorporated to support SOS messaging in such scenarios.

The system can also be extended with V2X communication, including Vehicle-to-Vehicle and Vehicle-to-Infrastructure protocols. This would allow a crashed vehicle to broadcast warning signals to nearby vehicles and road infrastructure, helping to prevent secondary accidents and reduce traffic congestion.

In addition, advanced sensor fusion can be introduced by integrating extra sensors such as ultrasonic sensors or specialized microphones. These sensors can help identify audio signatures like glass breakage or metal deformation, thereby improving accident detection confidence and overall system accuracy.

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